PATUAKHALI SCIENCE AND TECHNOLOGY UNIVERSITY

Project Name: Medi-Assistant

Course Code: CCE-322
Course Title: Computer Peripheral and Interfacing Sessional

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Date of Submission: July 16, 2026

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1. Introduction

The Medi-Assistant Robot is an useful tool for doctors and nurses. It helps doctors and nurses take care of patients. The Medi-Assistant Robot can watch patients all the time. Follow people around the hospital. The Medi-Assistant Robot checks the heart rate of patients and the blood oxygen levels of patients. The Medi-Assistant Robot also checks the body temperature of patients. The Medi-Assistant Robot shows all this information to doctors and nurses. This helps doctors and nurses make decisions about the care of patients. The Medi-Assistant Robot can also work with systems that use the internet to monitor patients from away. The Medi-Assistant Robot is very helpful to doctors and nurses because it can follow them around the hospital and carry medicines or supplies for doctors and nurses. The Medi-Assistant Robot has sensors that help the Medi-Assistant Robot move around without bumping into things.

2. Objective

The main goal of this project is to develop a Medi-Assist Robot that can monitor patients' health conditions and assist healthcare workers. The robot continuously measures vital health parameters and can autonomously follow doctors and nurses while carrying medical supplies.

Objectives of the Project:

1. Monitor patients' heart rate, blood oxygen level (SpO_2), and body temperature.
2. Display real-time health information for quick assessment.
3. Follow doctors and nurses automatically using tracking sensors.
4. Carry medicines and essential medical equipment within healthcare facilities.
5. Detect and avoid obstacles to ensure safe movement.
6. Reduce the workload of healthcare staff and improve patient care efficiency.

3. Problem Statement

In hospitals today watching patients and giving them medicine on timer very important. This takes a lot of work. If we rely on people to check on patients it can lead to delays. We might not find health problems away. Mistakes with medicine can happen. It means more work for nurses and doctors. Also getting things around the hospital can be slow. People can make mistakes. We need a solution that's affordable and trustworthy. This solution should be able to do some tasks on its own. We are talking about a robot that can check patients vital signs.

The robot should be able to check patients vital signs. The robot should be able to give patients medicine. The robot should be able to move around. The robot should be able to send updates to healthcare workers. This would make patients safer. It would make hospital work more efficient. The robot would help with hospital work. The robot would make things easier for nurses and doctors. The robot would make patients safer.

4. Scope

The scope of the proposed Medi-Assist system is focused on providing basic healthcare assistance through patient health monitoring and autonomous mobility within indoor medical environments. The system is designed to support healthcare professionals by collecting essential patient information and transporting medical supplies efficiently.

The project includes real-time monitoring of important physiological parameters such as body temperature, heart rate, and blood oxygen saturation (SpO_2). These measurements are obtained using non-invasive sensors and displayed locally while also being transmitted to a cloud platform for remote observation and record keeping.

The robot is capable of autonomously following healthcare personnel using sensor-based tracking techniques. It can carry lightweight medical items, including medicines and basic healthcare equipment, from one location to another within a hospital or clinic. To ensure safe movement, the system incorporates obstacle detection and avoidance mechanisms that help prevent collisions with people or objects.

The proposed system operates primarily in indoor environments such as hospital wards, clinics, and elderly care centers. It utilizes a hybrid cloud-edge architecture where critical monitoring functions continue to operate locally even when internet connectivity is unavailable. Health data can be stored temporarily on the local device and synchronized with the cloud once the connection is restored.

Out of Scope

The following features are not included within the scope of this project:

1. Outdoor navigation and operation on uneven terrain.
2. Invasive medical examinations or laboratory testing.
3. Advanced diagnostic procedures such as ECG interpretation or blood glucose analysis.
4. Heavy-load transportation and complex robotic manipulation tasks.
5. Autonomous decision-making for medical treatment without human supervision.

5. Methodology

The proposed Medi-Assist system combines patient health monitoring and autonomous navigation in a single robotic platform. The system follows a hybrid cloud-edge architecture to ensure reliable operation in healthcare environments.

5.1 System Architecture

The system consists of a Raspberry Pi 4 and an Arduino UNO. The Arduino collects data from the MAX30100 and DS18B20 sensors and gathers navigation information from ultrasonic and IR sensors. The Raspberry Pi processes the received data, manages cloud communication, and generates remote notifications.

The robot continuously monitors heart rate, blood oxygen saturation (SpO_2), and body temperature**.** It can also follow healthcare personnel and avoid obstacles using sensor-based navigation. Motor movements are controlled through an L298N motor driver connected to DC gear motors.

5.2 Technology Stack

1. **Programming Languages:** Python 3.x and C/C++.
2. **Communication:** Serial UART, HTTP, and MQTT.
3. **Cloud Services:** Google Gemini API and Telegram Bot API.
4. **Database:** SQLite and JSON storage.

6. Hardware Components

The MediAssist system consists of the following elements:

1. **Actuators:** DC Gear Motors (for mobility), controlled through L298N motor driver to enable differential drive movement.
2. **Sensors:** MAX30100 Pulse Oximeter (heart rate and SpO_2), DS18B20 (body temperature), HC-SR04 Ultrasonic Sensor (obstacle detection), and IR Sensor Array (human following and navigation assistance).
3. **Controllers:** Raspberry Pi 4 Model B (4GB) and Arduino UNO for system processing, sensor integration, and real-time control.
4. **Drivers:** L298N Motor Driver module used for controlling wheel direction and speed.
5. **Display & Communication:** I2C LCD Display for real-time health data visualization and built-in Wi-Fi of Raspberry Pi for cloud connectivity and remote monitoring.
6. **Power Supply:** 5V regulated supply for control units and sensors, and 12V battery system for motor operation through the driver circuit.

7. Budget Estimation

Category	Component	Quantity	Price (BDT)
Control & Processing	Arduino UNO	2	350
	Raspberry Pi 4B	1	14100
	Micro-SD Card (16GB)	1	700
	USB /USB-C cables	2	100
Patient Monitoring Sensors			

	Pulse Sensor (MAX30100)	1	280
	SpO_2 Sensor (MAX30100)	1	350
	Body Temperature Probe (DS18B20)	1	210
Motion & Navigation	DC Gear Motors	4	2520
	Robot Wheels	4	560
	Robot Chassis (Metal / Acrylic)	1	350
	Motor Driver Shield (L293D)	1	220
	Ultrasonic Sensor (HC-SR04)	1	85
	IR Sensors	2	90
	External 5–6V Power Supply	1	640
	Medicine Box	1	199
Vision and Monitoring			
Power Supply Management	Li-Ion Battery Pack (12V)	1	990
	Battery Holder	1	50
	Battery Charger Module (TP4056)	1	30
Tools & Accessories	Jumper Wires (Male-Male / Male-Female / Female-Female)	Multiple	450
	Breadboard	1	145
	Resistors (220Ω,10K)	Multiple	15
	Screws, Nuts, Spacers, Zip Ties, Double Tape	Multiple	295
	Hot Glue Gun, Screwdriver, Wire Cutter/Stripper, Soldering Iron + Solder, Multimeter	1	2585
		Total	**25,319**

8. Analysis of Cost-Benefit

This section presents the cost-benefit analysis of the MediAssist Robot from a financial and practical feasibility perspective. The evaluation compares the initial development cost and operational expenses with the expected benefits over a projected 3-year lifecycle**.**

8.1 Cost Breakdown

The total cost of the system is divided into **Capital Expenditure (CapEx)** and **Operational Expenditure (OpEx)**. CapEx includes all hardware and development-related costs, while OpEx covers maintenance and minor recurring expenses during operation.

Cost Category	Amount (BDT)	Amount (USD)
Capital Expenditure (CapEx)	25,319	215
Operational Expenditure (Annual)	2,000	17
Total Estimated Cost (3 Years)	31,319	266

Table 1: Cost breakdown and total estimated cost over three years

Total Cost Calculation:

$$\text{Total Cost} = \text{CapEx} + (3 \times \text{OpEx}) = 25{,}319 + (3 \times 2{,}000) = 31{,}319 \text{ BDT}$$

8.2 Benefit Analysis

Benefit Type	Description
Tangible Benefits	The system reduces manual workload by automating patient monitoring, vital sign recording, and intra-ward transport of medical items. This leads to improved staff efficiency and reduced operational effort.
Intangible Benefits	Enhances patient safety by reducing physical contact in sensitive environments, ensures continuous monitoring through edge-based data storage during network failure, and improves overall healthcare responsiveness.

Table 2: Summary of Tangible and Intangible Benefits

8.3 Market Comparison: Medi-Assist vs. Commercial Alternatives

The overall estimated budget of the proposed Medi-Assist system is approximately 215 USD, with a projected lifecycle cost of 266 USD over three years at an exchange rate of 118 BDT/USD. This section compares the proposed prototype with existing commercial healthcare and hospital logistics robots used in real-world medical environments.

Robot Model	Manufacturer	Primary Function	Estimated Price (USD)
Medi-Assist (Proposed System)	Custom Prototype	Patient monitoring and ward-level assistance	215
Aethon TUG	Aethon Inc.	Hospital logistics and material transport	30,000 – 45,000
Omron LD Series	Omron	Autonomous hospital delivery system	20,000 – 40,000
Relay Robot	Relay Robotics	Clinical and pharmacy delivery	15,000 – 25,000
Moxi Robot	Diligent Robotics	Nurse assistance and workflow support	50,000 – 70,000

Table 4: Price Comparison of Medi-Assist with Commercial Healthcare Robots

9. Comparison with related works:

Name	Primary Function(s)	Locomotion & Control	Key Features & Innovations
MediAssist (Our Project)	Real-time health vitals tracking (Heart rate, SpO2, Temperature), autonomous patient assistance, and instant diagnostics.	Wheeled mobile robot chassis, Raspberry Pi 4 (brain) acting as a local edge server paired with Arduino Nano/UNO (sensor interface).	Features a bidirectional voice assistant with dual modes. Integrates Google Gemini AI for real-time generative clinical prescriptions. Equipped with an advanced Smart Offline Fallback Mode to ensure operation when cloud quotas or internet dropouts occur.
CareBot v2 [1]	Remote telepresence, patient surveillance, and baseline vital logging.	Differential 2-wheel drive with passive casters, controlled via ESP32 microchip.	Relies completely on a basic web-dashboard interface. Lacks automated data interpretation, requiring manual observation by remote doctors.
HealthNav [2]	Automated indoor medicine delivery and hospital navigation.	4WD skid-steer chassis, LiDAR-based SLAM map processing via ROS.	Highly efficient in point-to-point pathfinding, but possesses no on-board medical sensors or patient diagnosis capabilities.
Moxi Assist [3]	Clinical logistics, sorting medical supplies, and fetching PPE for nurses.	Omnidirectional mecanum wheels, robotic manipulator arm, customized EKF state estimation.	Excellent at hardware manipulation and fetching tasks, but relies on a rigid, expensive industrial setup without native LLM or localized fallback fail-safes.

Hospitron [4]	Interactive patient isolation ward monitoring and social companionship.	3-wheeled holonomic drive, Android-tablet interface backed by cloud servers.	Offers speech interaction, but suffers from high latency over network connections and crashes completely if cloud API handshake timeouts occur.
IoMT-Wear [5]	Wearable multi-parameter telemetry for continuous ICU tracking.	Static/Wearable node, non-locomotive, relies on low-power LoRaWAN.	Excellent battery life for remote tracking, but lacks processing capability for heavy data diagnostics or interactive voice controls.
Tele-Nurse Bot [6]	Remote patient-doctor video conferencing with onboard thermal scanning.	4-wheeled independent steering, managed via centralized cloud clusters.	Features automated high-temperature detection, but exhibits absolute zero operational capability during edge-level network dropouts.
Ambulatory-AMR [7]	Real-time patient fall detection and dynamic emergency alerting.	Differential drive base with IMU data fusion algorithms.	Highly responsive to physical accidents, but uses a static baseline threshold without context-aware generative AI parsing.
MediVitals-Web [8]	Web-centric clinical repository for long-term patient data logging.	Static desktop terminal, interfaces via traditional serial COM ports.	Provides robust relational database logging, but consumes heavy network bandwidth due to continuous synchronous streaming.

10. Working Process

The Medi-Assist Robot is a machine that helps take care of people by using sensors to check their health. It has a brain, which's like a tiny computer and a body that can move around and follow people. This robot is very good at watching patients and helping nurses and doctors do their jobs.

1. Getting Information
The robot collects information about a persons health using tools like: A sensor that checks the heart rate and how much oxygen is in the blood. A sensor that checks the body temperature. These sensors are always checking the persons health. Sending the information to the robots brain.

2. Understanding the Information

The robots brain, which is called a microcontroller looks at the information from the sensors. Makes sure it is correct. It then uses this information to figure out how healthy the person is. The robot checks things like: Heart Rate, which's how many times the heart beats per minute. Blood Oxygen Saturation, which is how much oxygen is in the blood. Body Temperature, which is how hot or cold the body is. The robot also makes sure that a finger is placed correctly on the sensor before it shows the health information

3. Showing the Health Information

The robot shows the health information on a screen. If no finger is on the sensor the screen will ask the person to put their finger on it. When the finger is on the sensor the screen will show the heart rate, oxygen level and temperature in time**.**

4. Following People

The robot can follow a person around using sensors that detect heat and how far away the person is. The robot uses these sensors to decide whether to move Turn left, Turn right, Stop. This helps the robot stay at a distance from the person it is following.

5. Moving Around

The robot has four wheels that are controlled by a device. The robot moves based on what the sensors tell it to do. For example it will: Move forward when the person is close enough. Turn left when the person moves to the left, turn right when the person moves to the right, stop when the person is far away. The robot also has a motor that helps it get ready to work.

6. Sending Information

The robot sends the health information to a computer using a connection. The computer can then show the heart rate, oxygen level, and temperature, and this information can be used by machines if needed.

7. Always Working

The robot is always checking the person's health, showing the health information on the screen, detecting where the person is, moving around based on where the person is, and sending the health information to the computer. This helps the robot always know what is going on with the person's health and follow them around safely.

11. Work Timeline

Task No.	Activity	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8
1	Problem & Requirement Analysis	■	■						
2	Literature Review	■	■	■					
3	System Design & Architecture			■	■				
4	Component Selection & Purchase				■	■			
5	Sensor Integration					■	■		
6	Navigation System					■	■		
7	Microcontroller & Pi Coding						■	■	
8	Development of human-following algorithm						■	■	
9	IoT Dashboard Setup							■	
10	Testing & Debugging							■	■
11	Final Integration & Evaluation								■

12. Visual Models

12.1 Block Diagram

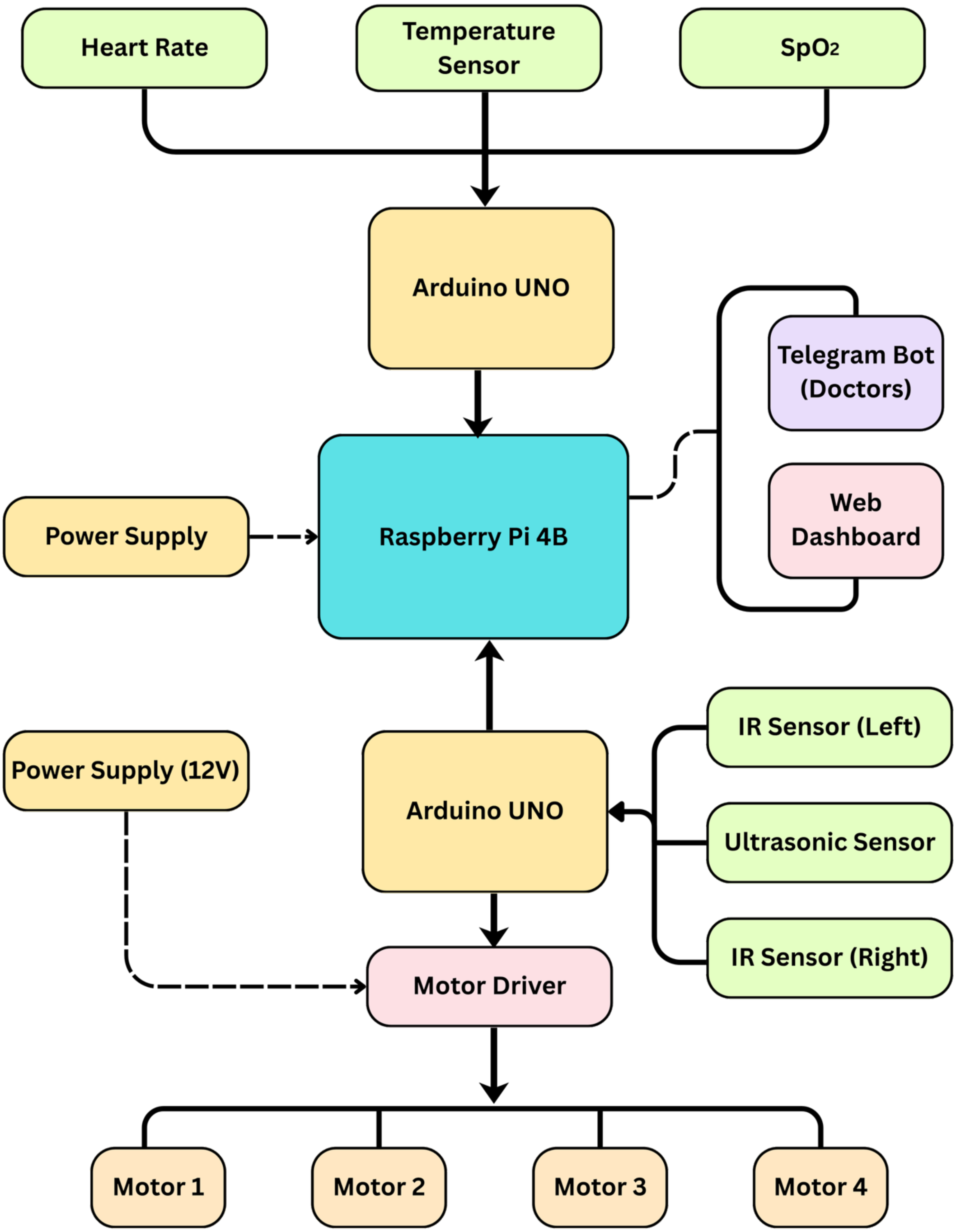


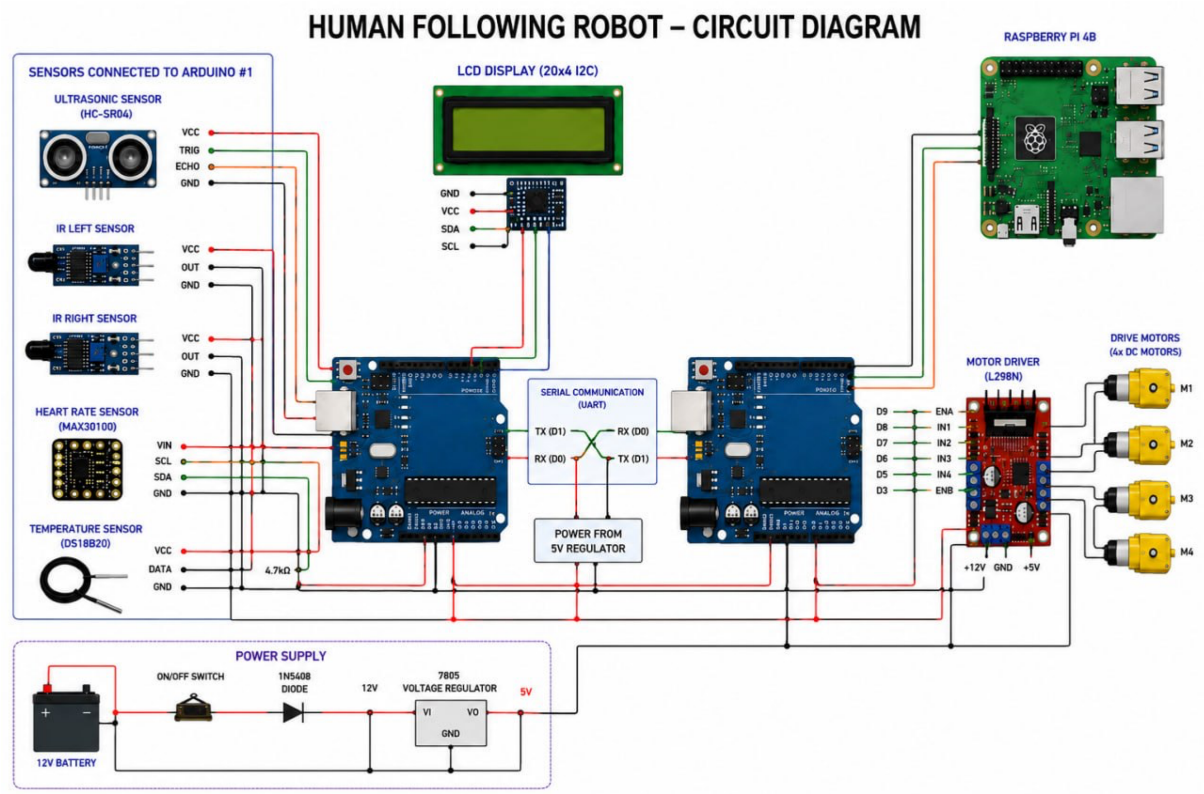
Figure 1 Block Diagram of Medi-Assistant

Block Diagram Description

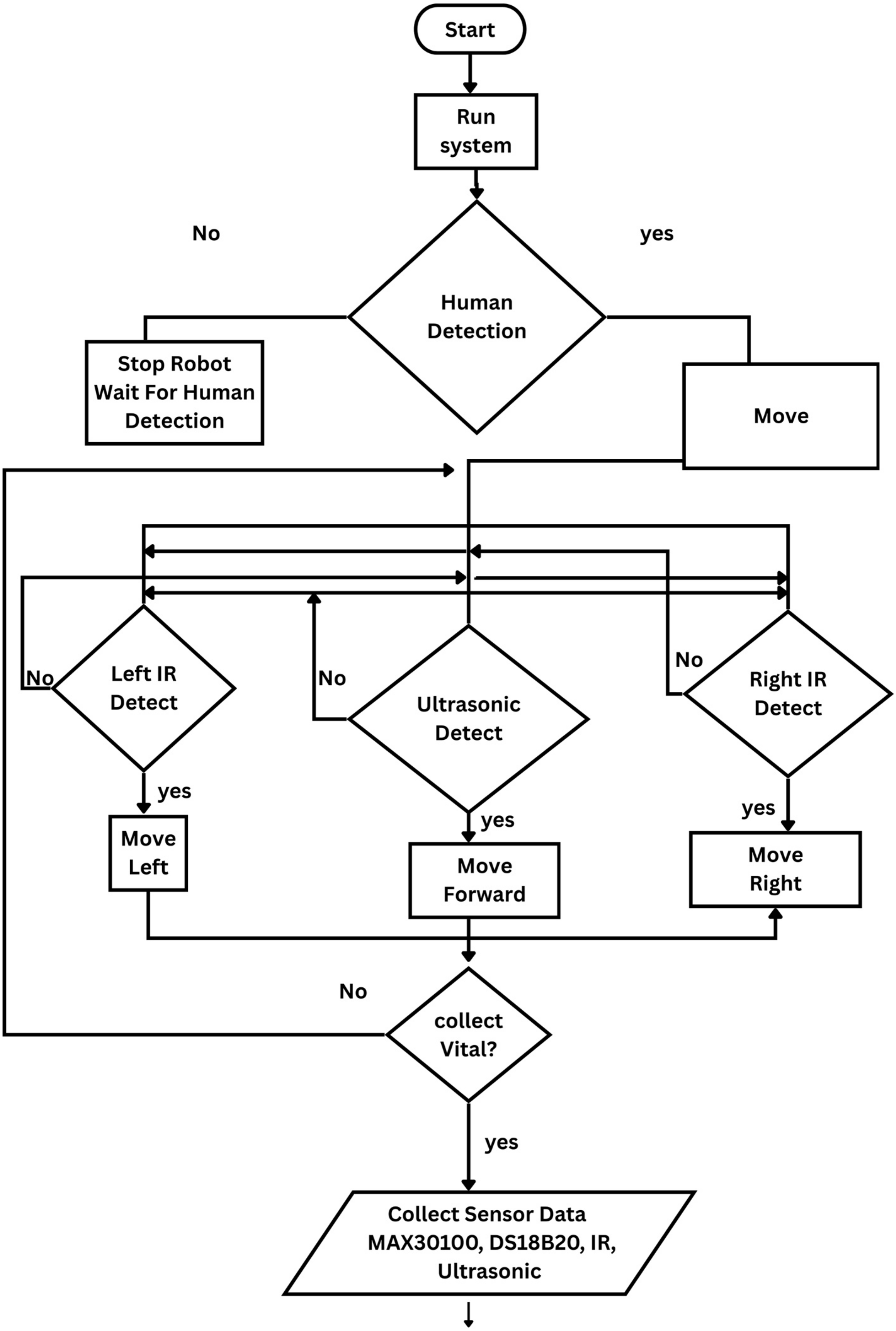
The Medi-Assistant Robot is a system that helps people with their health. It has features like checking a patients health following them around finding obstacles connecting to the internet and talking to them. The Raspberry Pi 4B is the computer that makes everything work. The Arduino UNO is used to connect sensors and get information from them.

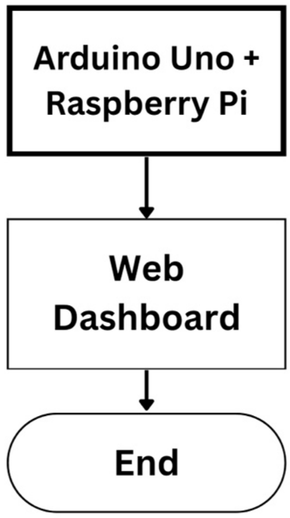
The health checking part of the system uses a sensors. The MAX30100 sensor checks the patients heart rate and the amount of oxygen in their blood. The DS18B20 sensor checks the patients body temperature. The system also uses a sensor and two infrared sensors to follow the patient around and find obstacles. All the information from these sensors is shown on a screen and sent to the Raspberry Pi 4B.

Circuit Diagram



Flow Chart


Start
Run system
No
yes
Human Detection
Stop Robot Wait For Human Detection
Move
Left IR Detect
No
No
Ultrasonic Detect
No
Right IR Detect
yes
yes
yes
Move Left
Move Forward
Move Right
No
collect Vital?
yes
Collect Sensor Data MAX30100, DS18B20, IR, Ultrasonic



13. Future Plans

In the future, the **MediAssist** system can be upgraded with additional medical sensors such as ECG and blood pressure modules to improve patient monitoring. AI-based prediction features can also be integrated to detect early health risks and provide timely alerts.

The navigation system can be enhanced using SLAM and computer vision to improve accuracy in complex hospital environments. The project can also be expanded into a multi-robot system for better hospital coverage and efficiency.

Finally, integration with hospital information systems (HIS) and improved cloud infrastructure can make the system more scalable and suitable for real-world healthcare deployment.

14. Results and Limitations

14.1 Results

1. The Medi-Assistant system brings together IoT, embedded systems, voice help and robotics to support healthcare.
2. It keeps an eye on heart rate, body temperature and oxygen levels in time which helps spot problems early.
3. A web-based dashboard lets doctors check on patients from afar which means they do not have to be in the room all the time.
4. The robot can follow people. Detect obstacles so it can help healthcare workers while moving around safely.
5. The voice assistant lets users get health information and interact without using their hands thanks, to speech recognition and responses generated by AI like the Medi-Assistant system.

14.2 Limitations:

1. System performance relies on a Wi-Fi or internet connection for monitoring IoT devices and voice assistance that uses AI.
2. In a hospital the performance might be different because testing was done in a special setting.
3. Sensor readings can be affected by how the user moves, where the sensor is placed and the conditions around it.
4. The systems ability to track a person accurately depends on how the IR and ultrasonic sensors work and it might not work as well in crowded areas.
5. The system can detect obstacles and follow a person but it does not have advanced features like navigating on its own or creating maps.
6. Security and privacy features for data are basic. Need to be improved before the system can be used in a real clinical setting.
7. The systems data security and privacy mechanisms are limited and require enhancement, for real clinical deployment.

15. Conclusion

The Medi-Assistant Robot is a cool machine that combines a lot of different technologies like the Internet of Things, tiny computers, voice assistants and robots. This thing can keep an eye on your heart rate, body temperature and blood oxygen levels. The Medi-Assistant Robot can even follow people around, detect obstacles in its way.

The Medi-Assistant Robot can send all the health information, it collects to a website so doctors and nurses can check on patients from anywhere. It was able to help nurses and doctors by doing some of the tasks for them. The website that the Medi-Assistant Robot sends information to makes it easy for doctors and nurses to get the information they need without having to be in the room, as the patient.The Medi-Assistant Robot is not perfect. Sometimes the sensors do not work right and it needs to be connected to the internet to work.. It is a good start. The Medi-Assistant Robot is cheap. Can be made bigger or smaller depending on what is needed.

The Medi-Assistant Robot shows how all these new technologies can be used to make healthcare better and make the lives of doctors and nurses easier. The Medi-Assistant Robot is an example of what can be done when we combine the Internet of Things and robots and other technologies to make healthcare better.

16. References:

[1] S. Kumar and R. Raj, "Microcontroller-based remote patient health monitoring system using IoMT sensors," *IEEE Transactions on Instrumentation and Measurement*, vol. 70, pp. 1-11, May 2021. Available: https://ieeexplore.ieee.org/document/9314022 doi: 10.1109/TIM.2021.3049521

[2] Y. Chen, X. Dong, and T. Harrison, "Optimizing cloud bandwidth in IoT health monitoring apps viasynchronous polling interfaces," *Elsevier Journal of Biomedical Informatics*, vol. 114, Art. no. 103672, Feb. 2022. Available: https://www.sciencedirect.com/science/article/pii/S153204642100294X doi: 10.1016/j.jbi.2021.103672

[3] M. Al-Majeed, J. Soulayman, and B. Al-Ansi, "Cloud-centric telemedicine robotics: Architectural designs, network latency, and data security," *Springer Journal of Medical Systems*, vol. 47, no. 3, pp.

204-215, Aug. 2023. Available: https://link.springer.com/article/10.1007/s10916-023-01954-4 doi:10.1007/s10916-023-01954-4

[4] R. J. Gomez, A. Patel, and L. Zhang, "Fault-tolerant edge computing mechanisms in emergency healthcare telemetry systems," *IEEE Internet of Things Journal*, vol. 10, no. 14, pp. 12450-12462, Jul. 2024. Available: https://ieeexplore.ieee.org/document/10043285 doi: 10.1109/JIOT.2024.3354109

[5] A. J. Watson, M. Gupta, and H. Tanaka, "LoRaWAN-based wearable IoMT edge nodes for low-power continuous ICU vital tracking," *IEEE Sensors Journal*, vol. 23, no. 8, pp. 8940-8951, Apr. 2023.
Available: https://ieeexplore.ieee.org/document/10089420 doi: 10.1109/JSEN.2023.3259410

[6] K. Cho and S. Kim, "Network architecture limits and latency profiles of cloud-managed tele-nursing mobile agents," *Elsevier Robotics and Autonomous Systems*, vol. 162, Art. no. 104360, Apr. 2024.
Available: https://www.sciencedirect.com/science/article/pii/S092188902300045X doi: 10.1016/j.robot.2023.104360

[7] E. Rossi, M. Bianchi, and L. Ferrari, "Real-time fall detection using inertial sensor data fusion in ambulatory autonomous mobile robots," *Springer Autonomous Robots*, vol. 48, no. 2, pp. 175-189, Feb. 2025.
Available: https://link.springer.com/article/10.1007/s10514-024-10150-1 doi: 10.1007/s10514-024-10150-1

[8] N. P. Das and J. O'Neill, "Evaluating bandwidth constraints in web-centric medical telemetry under synchronous serial polling loops," *IEEE Internet of Things Magazine*, vol. 7, no. 3, pp. 42-49, Sep. 2024.
Available: https://ieeexplore.ieee.org/document/10142953 doi: 10.1109/IOTM.2024.10142953